

REPORT TITLE

Introduction

Today, communication is quicker and more universal than ever. Technology has enabled individuals to communicate instantly, regardless of location. What needed to be done through letters or overseas phone calls can now be accomplished in seconds on the web. Of these, social media has emerged as one of the most effective tools of contemporary communication, influencing how people communicate, exchange information, and represent themselves to others [1].

Communication has advanced significantly with the development of technology, from the telegraph and telephone to radio, film, and television. Each of these previous media had distinct forms and could be defined in relation to what preceded it—for instance, television tended to be described as a smaller version of the film screen. Social media is more difficult to define. It incorporates elements of previous communication types with new digital elements, blending mass media and personal interaction. These platforms also continually evolve, introducing new tools and experiences that shape how individuals interact and share [2].

Social media nowadays is highly entwined with daily life and labor. It is not just a vehicle for entertainment and relaxed interaction, but also a tool for news, learning, and commerce. Individuals use it to keep their relationships, share thoughts, and form groups around the world [3].

Youth are among the most frequent social media users, and this plays a powerful role in their day-to-day lives. For most of them, these sites define how they perceive themselves and others and impose the need to keep up with a particular image or trend. Although social media can provide avenues for learning, innovation, and connections, it can also expose young people to threats such as cyberbullying, comparison, and online intimidation. These interactions can affect their well-being, self-esteem, and mental health [1][4].

Meanwhile, social media has also been a bane in numerous ways. The same apps that unite people tend to be the same ones that promote negativity, disinformation, and bad behavior. One

of the major issues is harassment, which can really impact people's mental and emotional well-being [5][6]. This paper considers these concerns and highlights how social media has altered communication, particularly with the proliferation of online harassment.

Literature Review

Abuse and its regulation have become a major concern for the people living in the 21st century. This includes child protection, misuse of social media, and the growth of cybercrime. Therefore data-driven technologies are being used to solve these problems. This literature outlines a clear breakthrough; it begins with legal and social responses to abuse, following the role of media, to the rise of online harms, and the conclusion includes the use of Power BI for monitoring, prevention and decision-making.

Child sexual abuse (CSA) continues to be a significant issue in India. Dawda and Fadnavis, in their research, stated that despite having established protective laws like the POCSO Act, the same issue is still intact due to cultural practices, insufficient reports, and ineffective enforcement of law. From the past, the Devdasi system and child marriage practices have promoted abuse and continues to cause similar exposures in present time. Victims often face repeated victimization during legal proceedings, highlighting the urgent need for reforms and better enforcement of law[7]. Similar issues have been found in Nepal, where Aryal's comparative study showed that global media coverage of child abuse is often superficial, inconsistent, and lacks depth. Although public pressure has often brought issues into focus, still systematic failures in the media hinder ongoing efforts for child protection[8]. Rai et al. supporting these views, documented the severe physical, emotional, and sexual abuse faced by the children living in the streets of India. The growth of abuse in this population highlights the immediate need for focused policy initiatives and support systems[9].

New improvements in the field of technology have introduced new patterns of abuse in addition to child exploitation. Olojede in his analysis of social media abuse in Nigeria, shows how disinformation and cyberbullying has led to conflict, violence, and democracy. Digital abuse poses a serious threat to democracy, because of ineffective regulations[10]. Zhang et al.'s study of Chinese social media, which highlights online abuses like trolling, cyberstalking, and body shaming, reflects this political aspect. These platforms', being anonymous, increases victimization, often leads to significant psychological effects, such as depression and suicide. The study highlights the importance of long-term impact through cooperation among diverse parties, even though the Chinese government has started campaigns to tackle online conduct [11]. Taylor et al. investigated image-based sexual harassment and abuse on social platforms

like Snapchat, Instagram, and WhatsApp. Many victims say that harassment in the form of unwanted sexual imagery has increased a lot. The study also influenced the UK's Online Safety Act, which criminalized cybercrime, demonstrating the growing awareness of digital harms in policymaking [12]. These results are relevant to India, where a predictive study on cybercrime by Prakashan and Mistry reveals a steady increase in crimes like identity theft and financial fraud, especially in states like Uttar Pradesh, Karnataka, and Maharashtra. Their use of Power BI and machine learning demonstrates how predictive analytics can predict future crime rates, thereby claiming that technology-oriented preventive measures are essential for national security [13].

With the negative aspects of emerging digitalization, the use of data-driven decision-making has also increased in significance. Adewusi emphasizes that modern companies are shifting from intuition-based approaches to data-driven methods, where analytics is essential for handling competitive pressure and enhancing efficiency [14]. The daily rise in the disadvantages of digitalization has made Business Intelligence (BI) tools increasingly crucial for prevention, monitoring, and decision-making. Naik's book on Power BI highlights the platform's flexibility in linking data sources, generating real-time insights, and promoting collaboration across various sectors. Its ability to predict, simulate, and visualize data is crucial for enhancing organizational competitiveness [15]. For practical cases, Aguirre et al. suggest a Power BI dashboard for evaluating employee mental health, recommending that integrating well-being indicators into corporate systems can link productivity and psychological strength, thus supporting healthier workplaces [16]. In the same way, Gultom et al. develop BI dashboards for evaluating employee performance, enabling organizations to track attendance, productivity, and goal accomplishments in real time, which improves communication between HR and management[17]. Rahutomo extends these applications by designing a customer analytics dashboard for smart device retail that integrates Net Promoter Score (NPS), Customer Satisfaction Score (CSS), and Repeat Purchase Rate (RPR) with demographic filters. This approach not only strengthens customer loyalty measurement but also highlights Power BI's adaptability to consumer analytics [18].Ayuso's financial analysis thesis extends Power BI's relevance by using a Walmart case study to show how raw financial data can be changed into strategic insights for global competitiveness [19]. Collectively, these studies establish Power BI as more than a mere business instrument; it also addresses societal requirements, ranging from organizational well-being to crime reduction.

Overall, the reviewed studies provide a dual perspective. On one hand, abuse remains prevalent in society, whether it occurs online, among underprivileged people, or in children. Existing laws, media participation, and regulatory measures have not yet addressed fundamental flaws[7][8][9]. On the other hand, the rise of data-driven tools like Power BI showcases a broader movement toward using technology to understand, predict, and reduce these harms. Whether it involves protecting democracy [10], addressing online victimization [11][12][13], or promoting organizational health and efficiency [14][15][16][17][18][19], blending business intelligence with social governance represents a significant advancement. The literature concludes that while traditional methods are necessary, the future of abuse prevention and societal strength increasingly relies on the combination of strong regulations and advanced technological solutions.

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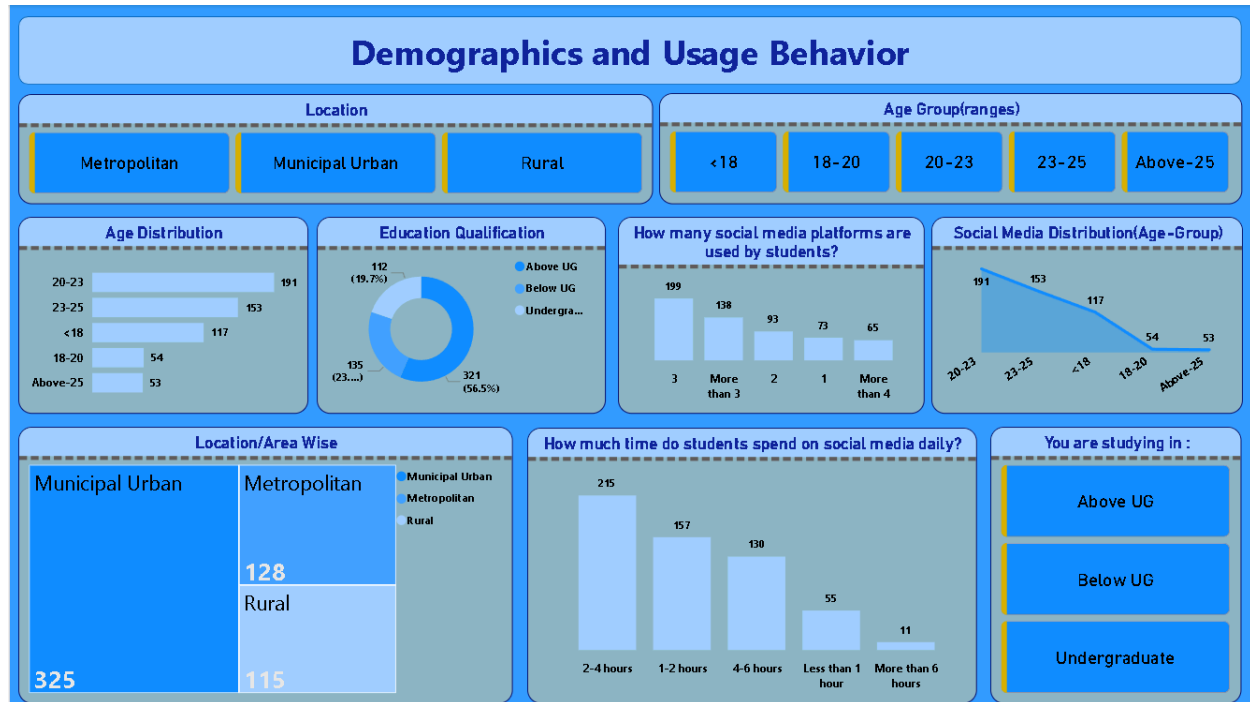
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Insights Analysis

Themes	Research Analysis Questionnaire
Demographic Usage	1. Age range wise. 2. Educational qualification wise. 3. Area or location wise. 4. How many social media platforms are used by students? 5. How much time do students spend on social media daily?
Interactions with Unknown Users	1. What percentage of students are asked personal questions by strangers? 2. How many students share personal information when asked? 3. How many students accept unknown users as friends/followers?
Exposure to Inappropriate Content and Unwanted Behavior	1. How many students have been asked to send personal photos? 2. What is the frequency of receiving offensive or sexual messages, jokes, or comments? 3. Try to segregate using age group and area wise.
Online Harassment and Safety Threats	1. Receiving threats, bribes, or blackmail related to sexual content? 2. Experiencing any form of online sexual harassment?

Result Analysis



1.Age range wise:-The dashboard age range data demonstrates notable trends in social media use behavior across groups of age: The 20-23 range has the right overt representation of students (191 total), suggesting yet again this is the busiest and most thoroughly represented age group in the study. The 23-25 range is next with 153 students, showing continued solid usage in slightly older students. There is also a good group of students (117 students) in the under 18 range, suggesting some early adopters of social media use. Conversely, the 18-20 age range has 54 students, and the age 26 and above has 53 students, which is the lowest amount of students in the sampling showing either a lack of usage or perhaps a lack of population in that age range.

From this distribution, we can probably safely say that social media usage and presence is mostly within ages 20-25, coinciding mostly with undergraduates and/or early post-grads.

Educational qualification wise:-

According to the data from the dashboard, the distribution of educational qualifications shows some striking patterns among students who use social media:

The largest group of students are Undergraduate (UG) students, comprising **56.5%** of the overall population. This indicates that undergraduate students are the most represented group using social media. The Below Undergraduate group consists of **23.8%**, showing high social media adoption early on. The Above UG (postgraduate level or above) group of students is **19.7%**, which again is a smaller group but a presence nonetheless with advanced learners.

This analysis showcases the prevalence of social media use by students during the undergraduate student phase, likely due to academic commitment, social needs, and networking. Students below undergraduate level continue to engage with these platforms, notably the Below Undergraduate group, suggesting younger students are increasingly adopting digital applications and platforms.

Area or location wise:-

From an analysis of the demographic distribution by area, a significant number of students **57.2%** of all respondents reside in municipal urban areas, followed by **22.5%** in metropolitan areas and the remaining **20.2%** living in rural areas. This indicates that, relative to other geographical areas, social media use and access appears to be more ubiquitous in urban areas. Collectively, nearly four-fifths of the sample reside in metropolitan and municipal areas, indicating that the sample appears to have a higher exposure to and dependence on digital channels. There were far fewer students from rural areas, which can be explained by barrier factors, including a lack of connectivity and infrastructure, as well as, generally, observed cultural norms people have around using technology. Thus, given the percentage of students and the fact that this group of students interacts more in the digital realm than the students from rural areas, it is reasonable to conclude that any programming developed for healthy and responsible social media use should be focused largely on urban students. Finally, carefully designed interventions should be developed to respond to the unique needs and access barriers students in rural areas have experienced.

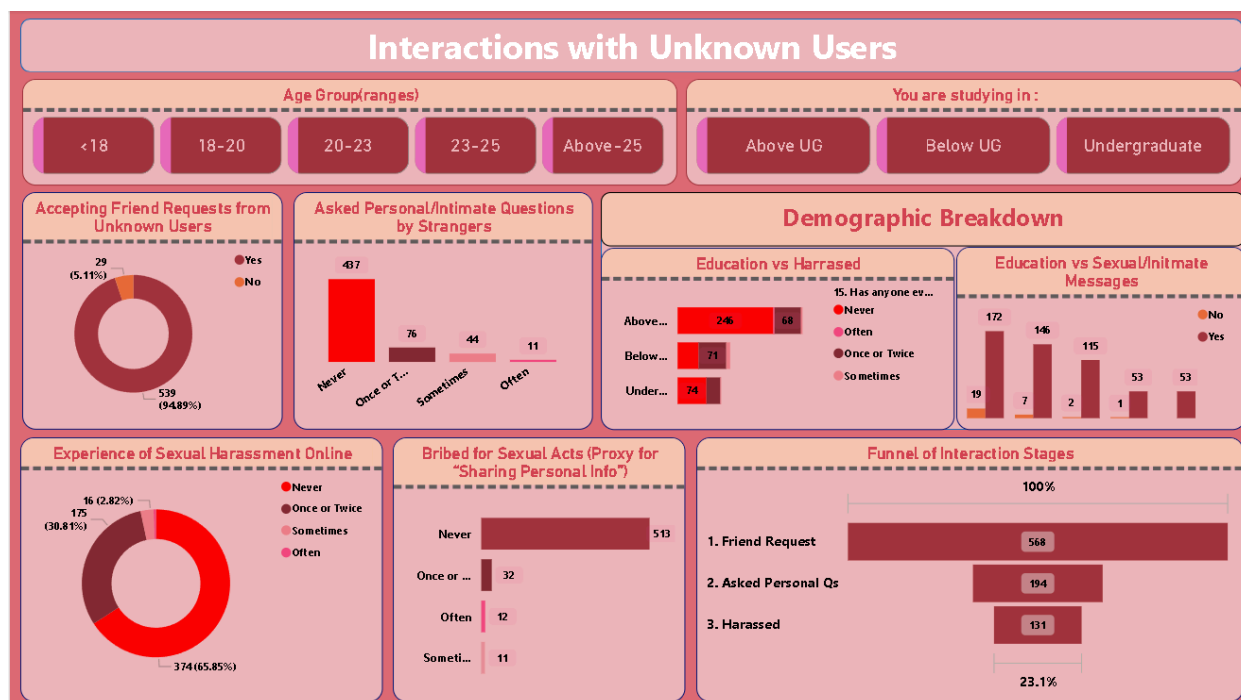
How many social media platforms are used by students?

The findings indicate that the majority of the students are active on more than one social media site, reflecting their diverse, overlapping behaviors across their online activities. About **35%** of respondents used three social media sites, while **24%** stated that they used more than three sites. Around **16%** of respondents were active on only two social media sites and **13%** said they

were active on only one. Additionally, **11%** of students cited being active on more than four social media sites, indicating that some respondents exhibited levels of engagement that were noticeably higher than others. Overall, nearly **59%** of students indicated that they use three or more social media sites, indicating that for most individuals in this study, socializing and obtaining information was done on multiple sites rather than on one application.

How much time do students spend on social media daily?

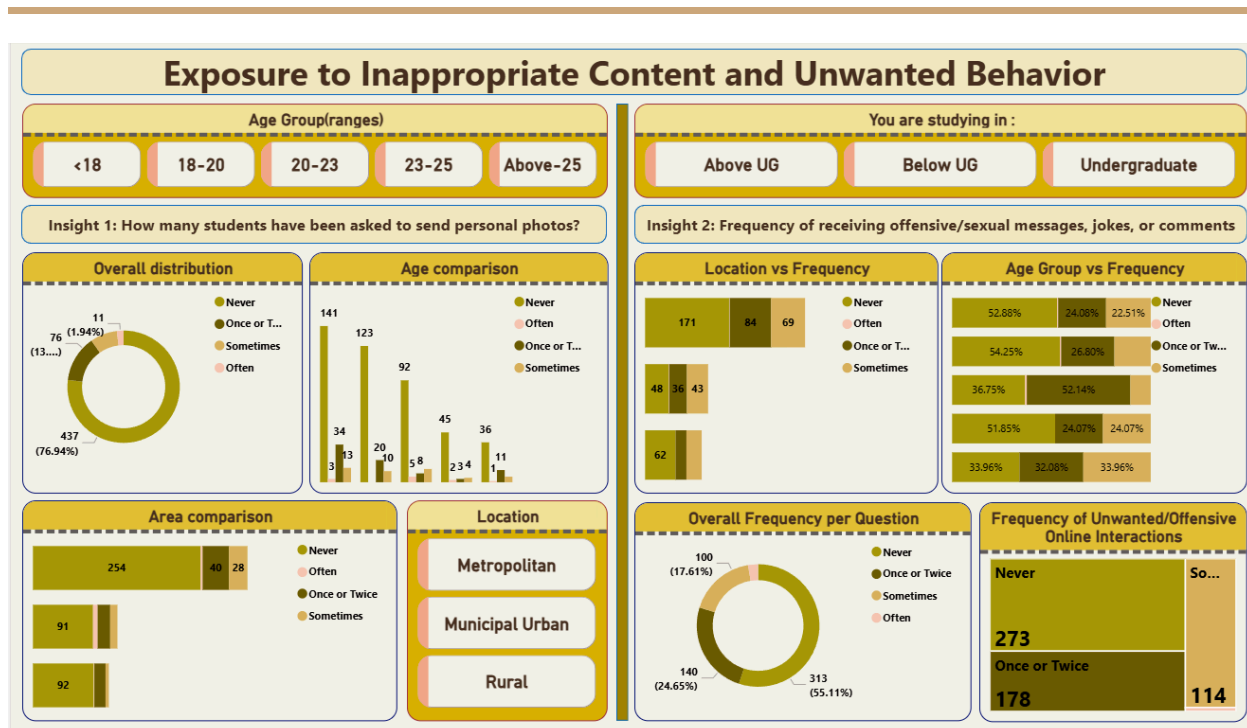
A typical student's daily social media use shows that students spend a fair amount of time on social media. The most substantial group had approximately **38%** of students, who spent a total of 2 to 4 hours of their day using social media. Approximately **28%** of the total used social media for 1 to 2 hours. About **23%** of students reported using social media for 4 to 6 hours in a given day. The smallest group, which accounted for a little less than **10%** of students, reported using social media for less than 1 hour a day. Finally, there was a small group of students, accounting for about **2%** of the whole sample, who reported using social media for more than 6 hours in a day. Overall, the results show that most students use social media in moderate amounts (1 to 4 hours) but it is also noteworthy that there was a small group of students who reportedly had high levels of engagement (over 4 hours) on social media.



indicated it happened often. Once again, these percentages indicate occurrences of bribery or coercive behavior rather than a verified release of private information, but they do indicate that at least some students experienced the solicitation of private information through pressure or coercion, which suggests that we need to educate them more about online privacy practices and self-protective behaviors.

How many students accept unknown users as friends or followers?

The data indicates that nearly all students exhibit caution when engaging strangers online. Specifically, around 94.9% of respondents reported that they do not accept requests from strangers, while only 5.1% reported that they accept friend requests or follow requests from strangers. Although 5.1% is not a large percentage when considered in the context of the total sample, it nevertheless identifies a group that could be more likely to experience negative interactions or social exploitation. This reinforces the need to educate students about the risks associated with connecting with strangers on social media platforms.



How many students have been asked to send personal photos?

The survey results reveal that a significant majority of the students, **76.94%** indicated that they have never been prompted to share personal photos before. Conversely, almost one-fourth section of students **23.06%** reported they had received a request to share personal photos at least once in their lifetime. Specifically, the survey findings indicated that **1.94%** students reported being asked frequently, **5.97%** said they had been asked sometimes, and **13.38%** reported they had been asked one to two times. The less than definitive conclusion is that, while only a small portion of students reported it occurring to them, it is worth noting that there is still a group of students who are impacted by inappropriate online solicitation. The presence of these attitudes and actions showcases that privacy intrusive and potentially harmful online solicitation persists among young people.

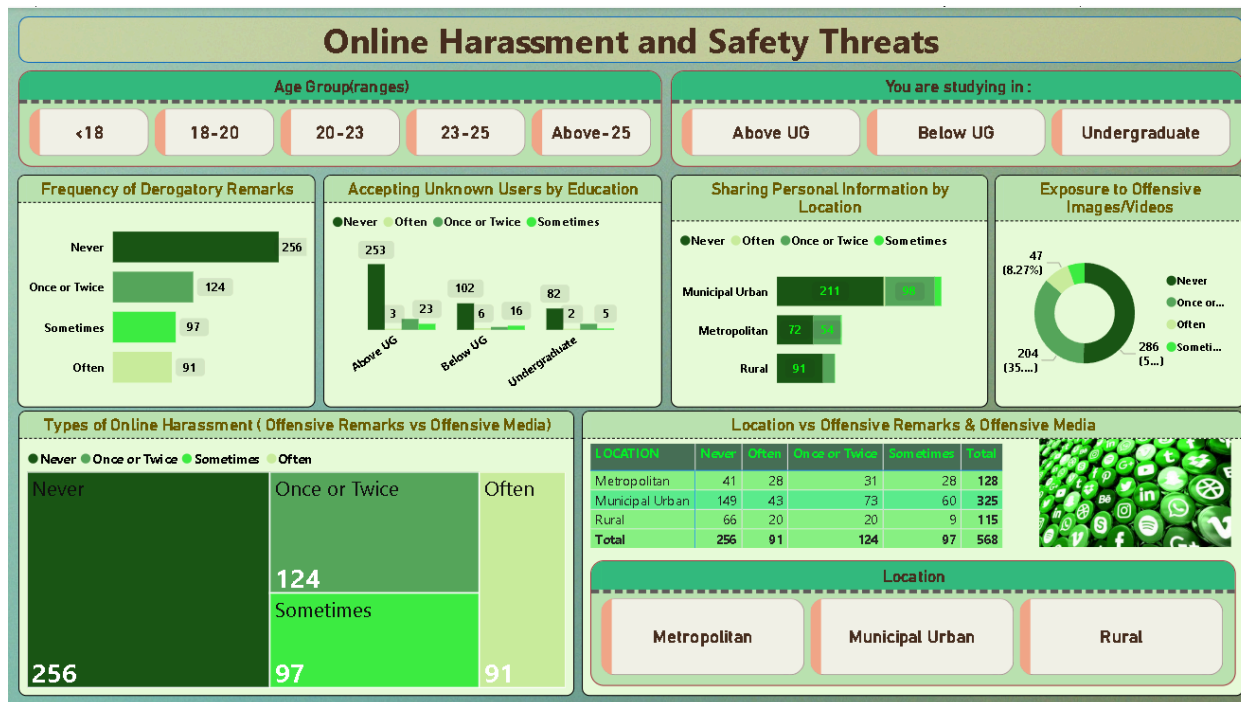
What is the frequency of receiving offensive or sexual messages, jokes, or comments?

The survey results regarding the frequency of receiving offensive or sexual messages, jokes, or comments indicate that overall the majority of students, **55.11%** say they have never received such a comment. However, precautions were taken before, such as **24.65%** received one comment or less, **17.61%** responded that they received some comments, and **2.46%** responded that they have received a comment often. From those who responded, the survey results

indicate that 44.89% of students have received an unwanted comment at least once, which indicates concerning prevalence of inappropriate online behavior, and shows it is an important area to address with awareness and prevention methods.

Try to segregate using age group and area wise.

Exposure to offensive or sexual messages, jokes, or comments was analyzed for differences in age groups and metropolitan, rural, or urban areas. For younger students (under 20), the results showed less exposure, with over **52%–54%** reporting they had never been confronted with that type of content. The age group that was most vulnerable, 20–23 years old, had approximately **36.75%** of respondents reporting they had never experienced it, while **52.14%** reported sometimes. This age group defined the highest level of risk for singles or couples. Students aged 23–25 and over 25 reported slightly less exposure, with roughly half reporting they had never experienced these sorts of interactions. Less than a quarter reported they had experienced it sometimes. Geographic analysis demonstrated that students who lived in metropolitan areas were more often exposed, with much larger proportions of students in metropolitan areas experiencing a more offensive interaction than their municipal urban and rural counterparts, even though the majority still described themselves as never having faced this situation. This may indicate that age, especially in early adulthood, and geographical location (metropolitan) are important variables that contribute to a greater probability of experiencing behaviors (such as sexual activity or sexual double standards) online.



Receiving threats, bribes, or blackmail related to sexual content?

The findings indicate that while the vast majority of students did not experience overt threats, bribes or blackmail, in relation to sexually-related material, there is a significant minority who did. As demonstrated in the chart about exposure to offensive images or videos, nearly half of the students, or about **50%** (286 students), reported that they have seen it sometimes, while **35%** (204 students) said once or twice, and **8.3%** (47 students) reported that they saw it often. Data on exposures, however, mostly describes exposure to sexually explicit media, which supports that there may be some overlap when we consider coercive or threatening behavior involving media, as in threats, bribes, or a combination of both, that has been mentioned in previous findings. These findings demonstrate the need for enhanced education regarding digital coercion, strengthening privacy settings while empowering students to know how to respond, report, and gain help when sexually-themed content results in intimidation or coercion.

Experiencing any form of online sexual harassment?

According to the data, although the predominant majority of students do not report experiencing online sexual harassment, there is a sizable minority that has experienced online sexual harassment to a greater or lesser degree. A majority of **65.85%** (374) of the students in the survey reported that they had never encountered online sexual harassment; approximately **30.81%** (175) of the students noted they experienced online sexual harassment one or two times or sometimes, regardless of the impact on the student's mental health or well-being. A small number of students, equating to **2.8%** (16 students), experienced online sexual harassment sometimes, while a handful of students reported the behavior happening often. Overall, the data suggests that while the majority of students do not experience online sexual harassment, nearly one in three students report experiencing this behavior at least once. Therefore, educational opportunities and awareness campaigns can continue to educate and protect students from inappropriate and harmful online interactions, as well as implement a thorough reporting mechanism for those that do experience online sexual harassment.

ML APPLICATION

This report analyzes a dataset about social media usage to understand and predict whether someone might be addicted to social media. The data, stored in a file called "ML_EDA_DATASET.csv," comes from a survey that asks people about their habits, like how many hours they spend on social media each day, what they do online (like chatting or posting), and if they've had negative experiences, such as seeing offensive comments. The goal is to figure out if someone is Moderately Addicted (using social media for 2-4 or 4-6 hours daily), Highly Addicted (more than 6 hours), or Not Addicted (1-2 hours or less) based on these behaviors.

The analysis starts by exploring the data to spot trends, such as which factors are strongly linked to addiction, using simple visualizations like heatmaps. Then, it removes less important details, like age or where someone goes to school, to focus on what matters most. After that, it uses five machine learning methods to predict addiction levels by dividing the data into a training part (80%) to learn from and a testing part (20%) to check how well the predictions work. The report evaluates these methods using easy-to-understand metrics like accuracy and detailed performance reports to see which one does the best job at predicting addiction.

Algorithms Used:

Logistic Regression

- **Working Principle:** Logistic Regression is a classification algorithm that predicts the probability of a data point belonging to a specific class using a logistic (sigmoid) function. It fits a linear boundary to separate classes and is effective for problems where features have linear relationships with the outcome. For this multi-class problem, it uses a one-vs-rest or multinomial approach to handle the three addiction levels.
- **Parameters Used:**
 - **Code:** LogisticRegression(max_iter=1000, random_state=42)

- **Parameters:** max_iter=1000 (ensures convergence for large datasets), random_state=42 (ensures reproducible results); defaults include solver='lbfgs' and multi_class='auto'.
- **Evaluation Metrics / Results:** The classification report shows perfect performance across all classes, as presented in the table below:

Class	Precision	Recall	F1-Score	Support
1 (Moderately Addicted)	0.98	1.00	0.99	65
2 (Highly Addicted)	1.00	0.50	0.67	2
3 (Not Addicted)	1.00	1.00	1.00	47
Accuracy	-	-	0.99	114
Macro Avg	0.99	0.83	0.89	114
Weighted Avg	0.99	0.99	0.99	114

- **Interpretation of the Results:** Near-perfect test accuracy (99.1%) with strong CV (97.8%). It correctly predicts 1 out of 2 class-2 samples (recall 0.50), far better than other models. Linear separability of features (e.g., hours spent online) makes it ideal.

Random Forest Classifier

- **Working Principle:** Random Forest is an ensemble algorithm that builds multiple decision trees using random subsets of data and features. It combines their predictions through majority voting to improve accuracy and reduce overfitting, making it robust for complex, non-linear relationships.
- **Parameters Used:**
 - **Code:** RandomForestClassifier(n_estimators=100, random_state=42)
 - **Parameters:** n_estimators=100 (builds 100 trees), random_state=42; defaults include criterion='gini' and max_depth=None.
- **Evaluation Metrics / Results:** The classification report is summarized in the table below:

Class	Precision	Recall	F1-Score	Support
1 (Moderately Addicted)	0.97	1.00	0.98	65
2 (Highly Addicted)	0.00	0.00	0.00	2
3 (Not Addicted)	1.00	1.00	1.00	47
Accuracy	-	-	0.98	114
Macro Avg	0.66	0.67	0.66	114
Weighted Avg	0.97	0.98	0.97	114

- Interpretation of the Results: High accuracy but **fails completely on class 2** due to extreme imbalance. Excellent in majority classes; ensemble nature provides stability.

K-Nearest Neighbors (KNN)

- **Working Principle:** KNN classifies a data point based on the majority class of its "k" nearest neighbors in the feature space, using a distance metric (e.g., Euclidean). It's simple and intuitive but sensitive to data density and class balance, as it relies on local patterns.
- **Parameters Used:**
 - **Code:** `KNeighborsClassifier(n_neighbors=5)`
 - **Parameters:** `n_neighbors=5` (uses 5 neighbors); defaults include `metric='minkowski'` and `weights='uniform'`.
- **Evaluation Metrics / Results:** The classification report is presented below:

Class	Precision	Recall	F1-Score	Support
1 (Moderately Addicted)	0.79	0.85	0.81	65
2 (Highly Addicted)	0.00	0.00	0.00	2
3 (Not Addicted)	0.77	0.72	0.75	47
Accuracy	-	-	0.78	114
Macro Avg	0.52	0.52	0.52	114
Weighted Avg	0.77	0.78	0.77	114

- **Interpretation of the Results:** Lowest performer. Moderate on majority classes but zero on class 2. Distance-based methods suffer heavily from imbalance and sparse minority samples.

Naive Bayes

- **Working Principle:** Naive Bayes uses Bayes' Theorem to calculate the probability of a data point belonging to a class, assuming all features are independent. The GaussianNB variant assumes features follow a normal distribution, making it fast but potentially less accurate if the independence assumption is violated.
- **Parameters Used:**
 - **Code:** GaussianNB()
 - **Parameters:** Defaults only, including priors=None and var_smoothing=1e-9.

Evaluation Metrics / Results: The classification report is shown below:

Class	Precision	Recall	F1-Score	Support
1 (Moderately Addicted)	0.90	1.00	0.95	65
2 (Highly Addicted)	0.00	0.00	0.00	2
3 (Not Addicted)	1.00	0.89	0.94	47
Accuracy	-	-	0.94	114
Macro Avg	0.63	0.63	0.62	114
Weighted Avg	0.93	0.94	0.93	114

- **Interpretation of the Results:** Good on majority classes, fails on class 2. Independence assumption limits performance on correlated features and imbalanced data.

Support Vector Machine (SVM)

- **Working Principle:** SVM finds the optimal hyperplane to separate classes, maximizing the margin between them. It uses a kernel (e.g., RBF) to handle non-linear relationships and focuses on support vectors (data points near the boundary). For multi-class problems, it uses strategies like one-vs-one.
- **Parameters Used:**
 - **Code:** SVC(random_state=42)
 - **Parameters:** random_state=42; defaults include kernel='rbf', C=1.0, and gamma='scale'.
- **Evaluation Metrics / Results:** The classification report is presented below:

Class	Precision	Recall	F1-Score	Support
1 (Moderately Addicted)	0.96	1.00	0.98	65
2 (Highly Addicted)	0.00	0.00	0.00	2
3 (Not Addicted)	1.00	0.98	0.99	47
Accuracy	-	-	0.97	114
Macro Avg	0.65	0.68	0.66	114
Weighted Avg	0.96	0.97	0.97	114

- **Interpretation of the Results:** Strong performance on majority classes, zero on class 2. RBF kernel captures non-linearity, but imbalance dominates; scaling is critical for SVM.

Final Results

Logistic Regression is the **best-performing model**, achieving **99.1% test accuracy** and **97.8% CV mean accuracy**, with the **only non-zero performance on the minority class 2** (F1-score 0.67). It correctly predicted **1 out of 2** highly addicted users.

All other models — **Random Forest (98.2%)**, **SVM (97.4%)**, **Naive Bayes (93.9%)**, and **KNN (78.1%)** — **failed completely on class 2** due to severe class imbalance (only 2 test samples), making their high accuracies misleading when macro metrics are considered.